

Md Fahim F. Chowdhury
Ph.D. Candidate
Graduate research and teaching assistant
Mechanical and nuclear engineering
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Education

- **Virginia Commonwealth University**, Richmond, Virginia, USA
 Department of Mechanical and Nuclear Engineering **Present**
Ph.D. candidate, Graduate research and teaching assistant

- **Auburn University**, Auburn, Alabama, USA **2020**
 Department of Electrical and Computer Engineering
Ph.D. student, Graduate research assistant
- **University of Alabama**, Tuscaloosa, Alabama, USA **2018**
 Department of Electrical and Computer Engineering
Ph.D. student, Graduate research assistant
- **Tuskegee University**, Tuskegee, Alabama, USA **2018**
 Department of Electrical Engineering
Master's student, Graduate research and teaching assistant
- **Bangladesh University of Engineering and Technology** **2013**
 Department of Electrical and Computer Engineering
Bachelor student in Electrical and Electronic engineering

Thesis and projects

- **Nano-oscillator based straintronic physical reservoir computing.** **Present**
 (Ph.D. dissertation), Virginia Commonwealth University, Richmond, Virginia
- **Coalitional Integration of Wind Turbines and Combined Emission Economic Dispatch in Power Systems** **2018**
 (Master's thesis), Tuskegee University, Tuskegee, Alabama
- **Study of Soft Computing Techniques for processing and feature extraction from electro-cardiogram (ECG)** **2013**
 (Bachelor thesis), Bangladesh Univ. of Engineering and Technology, Bangladesh
- **Design and implementation of Microcontroller based radio frequency transmitter-receiver using wireless controlled car** **2011**
 (Senior design project), Bangladesh Univ. of Engineering and Technology, Bangladesh

Research Experiences: Instrumentations, Techniques, and Roles

- Graduate research assistant, Virginia Commonwealth University, Richmond, VA.
 - Analysis and optimization of nanomagnets induced magnetic control field for quantum application.
 - High fidelity energy efficient quantum control of electron spin in mesoscopic qubit.
 - Design and analysis of spintronic physical reservoir computing with surface acoustic waves (SAW).
 - Design and analysis of skyrmion based energy efficient straintronic physical reservoir.
- Graduate research assistant, Auburn University, Auburn, AL.
 - Worked on design, simulation, analysis, fabrication, assembly, and testing of cable-to-cable micro-connector for superconducting thin-film flexible parallel transmission lines for quantum computer application. The flexible cables are constructed using multiple layers of Nb and polyimide. The assembly was validated and characterized from 293 K to 4 K for DC properties.
 - Worked on micro-fabrication of thin-film flexible membrane with Cu pillars/SU-8 pillars, 77 K to 4 K flexible micro-strip line cables.
 - Worked on process development for low pitch, high aspect ratio of micro-structures of SU-8 (an epoxy based negative photoresist) on silicon, fused silica, and molybdenum wafers.
 - Worked on designing, machining (several parts made by different vendors), and assembling a cryogenic test probe for DC and microwave characterization at low temperature (up to 4 K/LHe).
 - Worked on microwave simulation on multi-layer bridging connector, differential pairs in ANSYS HFSS; solid mechanics and low temperature analysis of microstructures in ANSYS Mechanical and COMSOL.
 - Worked on micro-machining of aluminum with LPKF PCB prototyping machine and femtosecond laser machine.
 - Worked on studying force, displacement, modulus, hardness, and compliance of flexible Cu pillar membrane using Hysitron nano-indenter 850.
 - High volume measurement and inspection:
 - 4-point resistance measurement
 - Capacitance measurement
 - Fault analysis (short, open, dimension mismatch, impurities etc.)
 - Time domain reflectometry (TDR)
 - Failure analysis of flexible cables (e.g. cracks at low temperature cycles)
- Graduate research assistant, University of Alabama, Tuscaloosa, AL.
 - Worked on designing a glass-integrated ferrite antenna to enable frequency modulation (FM) communication for automotive applications.
 - Worked on designing a shark fin integrated multi-input-multi-output (MIMO) dual band antenna for 5G and Wi-Max communication for vehicle telematics.
 - To incorporate beam-forming (BF) technology in 5G frequency band, I have worked on designing a micro-strip line based Butler Matrix phase shifting input feed network for antenna arrays using ANSYS HFSS and Keysight ADS.
 - Worked on designing a dual band antenna to enable 5G (28 GHz) and DSRC (Dedicated Short Range Communication, 5.8 GHz) communication in V2V (Vehicle to vehicle) and V2I (Vehicle to Infrastructure).
- Graduate research assistant, Tuskegee University, Tuskegee, AL.

- For efficient coordination of high penetration of intermittent renewable energy in power grid, we introduced a cooperative game algorithm in a micro-grid to optimize real power loss while maintaining grid stability using MATLAB/MATPOWER.
- Developed a new approach of coalition formation among the generators (traditional and wind) for optimal dispatch to reduce the total cost of generation. A unique objection function was formulated analyzed for combined emission economic dispatch incorporating wind cost and power losses and a Shapley value based cooperative algorithm applied for fair distribution of generations.

Teaching experience

Graduate Teaching Assistant, Virginia Commonwealth University, Richmond, Virginia Department of mechanical and nuclear engineering

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| 1. Dynamics and kinematics | Fall'22 |
| 2. Process & Systems dynamics | Spring'21, Spring'22 |
| 3. Engineering statics | Fall'20 |

Graduate Teaching Assistant, Tuskegee University, Tuskegee, Alabama Department of electrical engineering

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| 4. Python programming | Spring'18 |
| 5. Theory and experiment of AC circuits | Fall'16, Fall'17 |
| 6. Theory and experiment of DC circuits | Spring'16, Spring'17 |

Industry Experience

- March'2015 – December'2015: PLC automation of Raw Material Handling System (RMHS) with Danieli, Italy for AKS.
- January'2014 – February'2015: Erection and commissioning of 230KV//33KV//11KV sub-station with ABB, India.
- May'2013 – December'2013: Erection and commissioning of 110MVA Static Var Compensator (SVC) with ABB, Sweden in Abul Khair Steel Melting Limited (AKSML) as Technical Management Trainee.

Conference presentations and attendance

1. Quantum control of spin qubit ensembles with nanomagnets: Exchange coupling for high fidelity

Md. Fahim F. Chowdhury, Mohamad Niknam, Md Mahadi Rajib, Walid Al Misba, Robert N. Schwartz, **2022**
Kang L. Wang, Louis-S. Bouchard, Jayasimha Atulasimha
The 67th Annual Conference on Magnetism and Magnetic Materials

2. Focused Surface Acoustic Wave Induced Nano-oscillator Based Reservoir Computing

Md Fahim F Chowdhury, Walid Al Misba, Md Mahadi Rajib, Alexander J Edwards, Dhritiman Bhattacharya, **2022**
Mathew S Varghese, Joseph S Friedman, Jayasimha Atulasimha
Annual GOMACTech Conference

3. Focused Surface Acoustic Wave Induced Nano-oscillator Based Reservoir Computing

Md Fahim F Chowdhury, Walid Al Misba, Md Mahadi Rajib, Alexander J Edwards, Dhritiman Bhattacharya, Mathew S Varghese, Joseph S Friedman, Jayasimha Atulasimha
American Physical Society's March Meeting **2022**

4. Focused Surface Acoustic Wave Induced Nano-oscillator Based Reservoir Computing

Md Mahadi Rajib, Walid Al Misba, **Md Fahim F Chowdhury**, Muhammad Sabbir Alam, Jayasimha Atulasimha
American Physical Society's March Meeting **2022**

5. Focused Surface Acoustic Wave Induced Nano-oscillator Based Reservoir Computing

Md Fahim F Chowdhury, Walid Al Misba, Md Mahadi Rajib, Alexander J Edwards, Dhritiman Bhattacharya, Mathew S Varghese, Joseph S Friedman, Jayasimha Atulasimha
Joint MMM-Intermag conferece **2022**

6. Towards Cable-to-Cable Connectors for Flexible Thin-Film Superconducting Transmission Lines

Archit Shah, Vaibhav Gupta, Bhargav Yelamanchili, Sherman E Peek, **Md Fahim F Chowdhury**, Asmaul Smitha Rashid, Uday S Goteti, John A Sellers, David B Tuckerman, Michael C Hamilton
Applied Superconductivity Conference 2020 **2021**

7. Game Theoretical Energy Management of EV Fast Charging Station with V2G Capability

Himadry Shekhar Das, Shuhui Li, **Md Fahim F Chowdhury**, Tonmoy Ghosh
IEEE International Conference on Power and Energy (PECon2020) **2020**

8. A Dual Band Shark Fin Integrated Vehicle Antenna For 5G and Wi-Max Applications

Sabah Mahjabeen Sarwar, **Md Fahim F Chowdhury**, Himadry Shekhar Das
IEEE International Conference on Telecommunications and Photonics (ICTP) **2019**

9. A Glass-Integrated Ferrite FM Antenna for Vehicle Telematics

Md Fahim F. Chowdhury, Yang-Ki Hong, Hoyun Won, Woncheol Lee, Minyeong Choi
IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting **2019**

10. Distributed Combined Emission-Economic Dispatch via Coalitional Integration of Wind Turbines

Md Fahim F. Chowdhury, Mandoye Ndoye, Gregory V. Murphy
SoutheastCon **2019**

11. Coalitional integration of wind turbines via cooperative energy trading in distributed power system.

Md Fahim F. Chowdhury, Mandoye Ndoye, Zhao Lu, Gregory V. Murphy
IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT) **2018**

12. Coalitional integration of wind turbines via cooperative energy trading in distributed power system

Md Fahim F. Chowdhury, Mandoye Ndoye, Zhao Lu, Gregory V. Murphy
2017 industry conference and NSF/DOE site visit **2017**

List of research publications

Niknam, M., **Chowdhury, M., Fahim, F.**, Rajib, M.M., Misba, W.A., Schwartz, R.N., Wang, K.L., Atulasimha, J. and Bouchard, L.S., 2022. Quantum control of spin qubits using nanomagnets. Communications Physics, 5(1), pp.1-10. **2022**

- Rajib, M.M., Al Misba, W., **Chowdhury, M.F.F.**, Alam, M.S. and Atulasimha, J., 2022. Skyrmion based energy-efficient straintronic physical reservoir computing. *Neuromorphic Computing and Engineering*. **2022**
- Chowdhury, M.F.F.**, Misba, W.A., Rajib, M.M., Edwards, A.J., Bhattacharya, D., Varghese, M.S., Friedman, J.S. and Atulasimha, J., 2022. Focused surface acoustic wave induced nano-oscillator based reservoir computing. *Applied Physics Letters*, 121(10), p.102402. **2022**
- Shah, A., Gupta, V., Yelamanchili, B., Peek, S. E., **Chowdhury, M. F. F.**, Rashid, A. S., ... & Hamilton, M. C. (2021). Towards cable-to-cable connectors for flexible thin-film superconducting transmission lines. *IEEE Transactions on Applied Superconductivity*, 31(5), 1-6. **2021**
- H. S. Das, S. Li, **M. F. F. Chowdhury**, T. Ghosh, “Game Theoretical Energy Management of EV Fast Charging Station with V2G Capability,” to be presented at 2020 IEEE International Conference on Power and Energy (PECon2020), virtual, Fall, 2020. **2020**
- Sarwar, S. M., **Chowdhury, M. F. F.**, & Das, H. S. (2019, December). A Dual Band Shark Fin Integrated Vehicle Antenna For 5G and Wi-Max Applications. In **2019 IEEE International Conference on Telecommunications and Photonics (ICTP)** (pp. 1-4). IEEE. **2019**
- Chowdhury, M. F. F.**, Hong, Y. K., Won, H., Lee, W., & Choi, M. (2019, July). A Glass-Integrated Ferrite FM Antenna for Vehicle Telematics. In **2019 IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting** (pp. 921-922). IEEE. **2019**
- Chowdhury, F. F.**, Ndoeye, M., & Murphy, G. V. (2019, April). Distributed Combined Emission-Economic Dispatch via Coalitional Integration of Wind Turbines. In **2019 SoutheastCon** (pp. 1-6). IEEE. **2019**
- Chowdhury, M. F.**, Ndoeye, M., Lu, Z., & Murphy, G. V. (2018, February). Coalitional integration of wind turbines via cooperative energy trading in distributed power system. In **2018 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT)** (pp. 1-5). IEEE. **2018**

Book chapters

- Das Himadry Shekhar, **Md Fahim F Chowdhury**, Shuhui Li, Chee Wei Tan. Fuel cell and hydrogen power plants. *Hybrid Renewable Energy Systems and Microgrids* 2021:313e49. <https://doi.org/10.1016/B978-0-12-821724-5.00009-X>. **2021**

Research funding agencies

- **Department of defense (DOE)**
- **National science foundation (NSF)**
- **Microsoft corporation**
- **Commonwealth cyber initiative (CCI)**

Software Skills

Programming Language

- MATLAB
- Python

Finite element method simulation

- ANSYS high frequency structural simulator (HFSS)
- ANSYS Mechanical

- Wolfram mathematica

- COMSOL Multiphysics

Micromagnetic simulation

- MuMax3

Circuit simulation

- Advanced design system (ADS)
- Cadence PSpice

Industrial automation

- Siemens Simatic Manager Step 7:
Programmable logic controller (PLC)

3-D Drawing and simulation

- SolidWorks 3D CAD

Languages

- **Bengali:** Native.
- **English:** Fluent.
- **Spanish:** Intermediate.
- **German:** Basic.

Awards and Fellowships

- IEEE MTT-S/IMS 2019 PhD Student Sponsorship Initiative Recipient.
- Engineering fellow, Government of Bangladesh, 2008-2012.

Professional memberships and Extra-curricular activities

- STEM outreach coordinator, IEEE Richmond section
- Treasurer at IEEE Magnetics society, student branch at VCU.
- Treasurer at Engineering graduate student association (EGSA) at VCU.
- Founder, Mind Shaper (micro-library project in rural areas of Bangladesh)
- Student ambassador of Trade Commission-Embassy of Spain.
- President at Tau Beta Zeta (engineering honor society) at Tuskegee University.
- Reviewer, SUSTECH 2022
- Reviewer, SoutheastCon 2022
- Reviewer, SoutheastCon 2023
- Session chair young professional at Southeastcon2021
- Student volunteer at International Microwave Symposium (IMS), Boston MA, 2019.
- Member at Institute of Electrical and Electronic Engineers (IEEE).
- Member of IEEE microwave theory and techniques society (IEEE-MTTS).
- Member at IEEE council on superconductivity, IEEE council on RFID, IEEE council on electronic design automation.
- Member of American physical society (APS)
- Volunteer at Tuskegee Feed-In program. Deborah Cannon Wolfe School
- Volunteer at HandsOn community service (Toys for tots).

List of collaborators: Past and present

Jayasimha Atulasimha, Va Commonwealth Univ.
 Louis-S Bouchard, Univ. of California LA
 Robert N Schwartz, Univ. of California LA
 Kang L. Wang, Univ. of California LA
 Mohamad Niknam, University of California LA
 Dhritiman Bhattacharya, Georgetown University
 Alexander J Edwards, Univ. of Texas at Dallas
 Joseph S Friedman, Univ. of Texas at Dallas
 Mathew S Varghese, Univ. of Texas at Dallas
 Md Mahadi Rajib, Va Commonwealth Univ.
 Walid Al Misba, Va Commonwealth Univ.
 Sabbir Alam, Va Commonwealth Univ.
 Sherman Peek, Auburn University
 Asmaul Smitha Rashid, Penn State University
 Uday S. Goteti, Univ. of California San Diego
 John A. Sellers, Auburn University

David Tuckerman, Microsoft corporation
 Michael C. Hamilton, Auburn University
 Archit Shah, Auburn University
 Vaibhav Gupta, Auburn University
 Bhargav Yelamanchilli, Auburn University
 Yang-Ki Hong, University of Alabama
 Hoyun Won, University of Alabama
 Woncheol Lee, University of Alabama
 Minyeong Choi, University of Alabama
 Himadry S. Das, University of Alabama
 Shuhui Li, University of Alabama
 Chee Wei Tan, Univ. Teknologi Malaysia
 Tonmoy Ghosh, University of Alabama
 Mandoye Ndoeye, Tuskegee University
 Zhao Lu, Tuskegee University
 Gregory V. Murphy, Tuskegee University

List of major course works at universities

Bangladesh University of Engineering and Technology

- Computer programming
- Energy conversion
- Engineering Electromagnetics
- Probability and statistics
- Linear algebra
- Ordinary and partial diff. equations
- Digital electronics
- Power system I & II
- Electrical properties of materials
- Communication theory
- Digital signal processing I & II
- Microprocessor and interfacing
- Random signals and processes
- Control system
- Solid state devices
- Microwave engineering
- Optical fiber communication
- Biomedical instrumentation
- Mobile cellular communication
- Industrial management
- Power system protection
- Power system operation and control
- Control systems

Tuskegee University

- Mathematical analysis
- Power system analysis
- Introduction to game theory
- Complex analysis for engineers
- Renewable energy
- Power electronics
- Computer network design

University of Alabama

- Integrated circuit fabrication
- Electric machinery

Auburn University

- Designing antennas and systems
- Semiconductor fundamentals
- Advanced quantum systems
- Modeling and simulation of multidisciplinary systems
- Photovoltaics

Virginia Commonwealth University

- Topics in nuclear engineering
- Neural nets and deep learning
- Heat and mass transfer theory
- Mech & Nuclear eng. Materials
- Nanomagnetic materials

